

Simir Birihan Atanaw

📍 Swindon, UK, ☎ +447592589461 | ✉ simirbirhan80@gmail.com | [LinkedIn](#) | [ORCID](#) | [ResearchGate](#)

Profile Summary

PhD-qualified Hydrologist with 7+ years of experience across hydrological modelling, hydrometry, catchment and water-resources management, field hydrology, flood and drought assessment, climate-impact analysis and environmental monitoring. Experienced in establishing and managing experimental catchment and river-flow monitoring activities, conducting field observations and inspections, supporting monitoring infrastructure and analysing hydrometeorological and environmental datasets for quality, accuracy and trends. Strong practical and analytical capability across flow monitoring, flow measurement, hydrological data interpretation, catchment assessment and environmental monitoring, with experience coordinating field activities, procurement and technical project deliverables. Advanced technical skills in SWAT+, HEC-HMS, ArcGIS, QGIS, R, Python, remote sensing and time-series analysis, complemented by strong technical reporting, stakeholder engagement and multidisciplinary project delivery experience. Experienced in flood and drought risk assessment, water resources management and climate adaptation, with a PhD in Hydrology & Water Resources Management and MSc in Engineering Hydrology. Full right to work in the UK, no sponsorship required, and available for relocation and frequent field-based site travel.

Key Achievements

- Supervised MSc **theses** and BSc **project work** to successful completion
- Managed international project **sustainable water resource management under pressure (SWAMP)**
- Published **peer-reviewed articles** in hydrology, modelling and water resources
- Developed **two MSc curricula** and revised a BSc curriculum in hydrology and water resources
- Established an **experimental catchment** for field based hydrological monitoring

Interested Role

Target roles such as Hydrologist, Flood & Drought Risk Analyst, Catchment manager, and Water Resources Project management within the UK. Interests include delivering practical research-based solutions for water resource planning and catchment management.

Professional Experience

Relocation August 2025- Mar 2026

- Moved from Ethiopia to the UK and reestablished career foundations.

Assistant Professor - Hydrology & Water Resources Management

University of Gondar, Ethiopia | Feb 2025 - Aug 2025

- Established and managed an experimental catchment for long-term hydrological monitoring, coordinating rainfall and river-flow observations, monitoring infrastructure and environmental data collection.
- Applied hydrological modelling and statistical analysis to assess catchment runoff, sediment dynamics and climate impacts, supporting peer-reviewed research and water-resources decision-making.
- Led multidisciplinary water-resources projects involving field monitoring, procurement, technical reporting, stakeholder engagement and international research collaboration.

Project manager - Sustainable Water Resources Management Under Pressure (SWAMP)|2024-2025

- Coordinated multidisciplinary water-resources research involving hydrological monitoring, field investigations and environmental datasets.
- Managed project activities, procurement, technical workshops, monitoring installations and delivery of project outputs.
- Coordinated collaboration between Ethiopian and international research partners.
- Prepared technical reports and supported stakeholder engagement and project communication.

Lecturer - Hydraulic & Water Resources Engineering |2017-2019

- Conducted field hydrometry and river-flow monitoring, including environmental observations, hydrological measurements and collection of data for catchment and water-resources assessment.
- Analysed and interpreted flow and environmental monitoring datasets, applying quality checks, time-series analysis and hydrological techniques to support technical assessments and reporting.

Assistant Lecturer - Water Resources & Irrigation Management |2015-2017

- Delivered tutorials in farm machinery, hydrometry, and laboratory analysis
- Delivered fieldwork and student research projects

Technical Skills

- **Hydrological Modelling:** SWAT+, HEC-HMS, calibration, validation, and uncertainty simulation
- **GIS & Remote Sensing:** ArcGIS, QGIS, and Google Earth Engine
- **Data Analysis:** R, Python, SPSS, Excel, statistical, and spatiotemporal modelling
- **Climate Impact Analysis:** Flood & drought, sediment, nutrient, and water-quality assessment
- **Field Hydrology:** Hydrometric measurements, laboratory analysis, and environmental monitoring
- **Project Delivery:** Leadership, stakeholder engagement, reporting, and communication
- Stakeholder engagement, teamwork, workshop & conference participation, and technical reporting

Developing skills

- UK Professional Development: FEH flood estimation, ReFH2 and HEC-RAS – currently developing through structured training and portfolio projects

Education

PhD - Hydrology & Water Resources Management

Africa Centre of Excellence for Water Management (ACEWM), Addis Ababa University |2019-2025

Thesis: *Modelling spatial patterns of runoff-generating areas and soil erosion in the Dirima Watershed, Lake Tana Basin using SWAT+ model.*

MSc - Engineering Hydrology

Bahir Dar Institute of Technology, Bahir Dar University |2015-2017

Thesis: *Assessment of hydrological responses in treated and untreated catchments.*

BSc - Water Resources & Irrigation Management

Wollega University |2011-2014

Thesis: *Evaluation of irrigation water demand for different vegetables using Cropwat 8.0.*

Professional Membership

- BHS – Individual Member, British Hydrological Society
- IAHS – International Association of Hydrological Sciences
- IWRA – International Water Resources Association
- (WRPA) Water Resources Professionals Association, Ethiopia (joint founder)

Selected Publications

1. Atanaw, S. B., Zimale, F. A., & Ayenew, T. (2024). ***Assessment of erosion, sediment yield, and runoff generating areas in Dirima catchment, upper Blue Nile, Tana Basin, Ethiopia.*** <https://doi.org/10.1007/s40899-024-01167-3>.
2. Atanaw, S. B., Zimale, F. A., Ayenew, T., & Ayele, G. T. (2023). Modeling future hydrological responses through parameter optimization and climate change scenarios in Dirima Watershed, Ethiopia. *Modeling Earth Systems and Environment*, 10(1), 1117–1135. <https://doi.org/10.1007/s40808-023-01817-z>.
3. Atanaw, S. B., Aynalem, D. W., Adem, A. A., Mekuria, W., & Tilahun, S. A. (2019). Effects of soil and water conservation practices on runoff, soil and nutrient losses in Alekt Wenz Watershed, Ethiopian Highland. *Lecture Notes of the Institute for Computer Sciences, Social-Informatics and Telecommunications Engineering, LNICST*, 274, 170–182. https://doi.org/10.1007/978-3-030-15357-1_14.

Certifications/ Professional Memberships

- Environmental Education: Climate Change -The Science (2026)
- Project Management in practice (2026)
- Nature Based Solutions for disaster and climate resilience (2025)
- Workshops in Kiel University Germany (2023 & 2025) for 15 days each year
- PhD research stays for one month in Kiel University, Germany (2022 & 2024)
- Research writing in the science and basics of grant writing (2022)
- ICAST conference participation (2018)
- Effective teaching training in University of Gondar (2016)

Additional Information

- Open to relocation for the right opportunity
- Available to travel as required

References

1. **Prof. Seifu A. Tilahun:** MSc. thesis Supervisor and teacher:
Senior Researcher-Hydrology and WR at International Water Management Institute
Email: satadm86@gmail.com | Phone: +251 97 850 1750
2. **Dr. Fasikaw Atanaw (Associate Professor):** teacher and PhD Supervisor:
Bahir Dar institute of technology PhD programs coordinator, and V/dean School of Graduate Studies
Email: fasikaw@gmail.com | phone: +251 91 870 1064
3. **Prof. Dr. Nicola Forher:**
Director of the Institute of Natural Resource Conservation, Department of Hydrology and Water Resources Management in CAU Kiel, Olshausenstr. 75
Email: nfohrer@hydrology.uni-kiel.de | Phone: (+49) 4318801276